DATAFACTZ · AI ENGINEERING INTERNSHIP

# Breach Analytics Platform

Use Case 3 capstone — agentic PII discovery, entity resolution, and exposure reporting over a synthetic document corpus

[Your name]

Week 4 · Capstone Presentation · [Date]

WHAT WE'LL COVER

# Agenda

20 minutes presentation, 10 minutes Q&A

**01**

### The journey

UC1 & UC2 in one line, then into UC3

**02**

### Problem & architecture

Pipeline vs. agent boundary

**03**

### Live demo

Ingest to exposure table and review

**04**

### Forced failure

The budget gate, caught and recovered

**05**

### Accuracy

Measured against planted ground truth

**06**

### Cost engineering

Tiered routing and the cost curve

THE PROGRAM

# Three weeks, three systems

UC1 · WEEK 1

## RAG Knowledge Chatbot

Azure stack. Retrieval-augmented answers over a knowledge base.

UC2 · WEEK 2

## Document Extraction

Open-source stack. Human-in-the-loop structured extraction.

UC3 · WEEKS 3-4

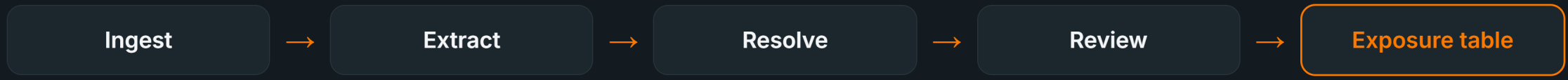
## Breach Analytics

This capstone. Agentic architecture, entity resolution, PII exposure reporting.

THE PROBLEM

# From a messy document pile to a reviewable exposure table

A heterogeneous corpus — emails, PDFs, scans, spreadsheets, Word docs — hides personal data in prose, tables, signatures, and images. The platform finds it, resolves it to real individuals, and makes every flag traceable to its source.



ARCHITECTURE

# Deterministic pipeline, agents only where judgment is needed

### PIPELINE — fast, cheap, testable

- Multi-format parsing + OCR
- Deterministic detectors (regex, Luhn checksums)
- LLM extraction validated via Pydantic
- Tiered model routing by document difficulty

### AGENTS — judgment & iteration

- **Orchestrator** — plans the campaign, adapts to failures
- **Exception investigator** — resolves or escalates quarantined files
- **Entity-resolution adjudicator** — reasons over uncertain matches
- **QA auditor** — independently re-verifies flags against source

Guardrails on every run: hard token/spend/step budgets, human approval gates on consequential actions, full inspectable run traces.

GROUND TRUTH

# A synthetic corpus, deliberately booby-trapped

**[500+]**
documents

**[150+]**
unique individuals

**[6]**
file types

**100%**
synthetic, zero real PII

**Edge cases planted on purpose**

Name variants (nicknames, maiden names, misspellings) · shared names across different people · identifiers split across documents · multi-person spreadsheets · false-positive traps (order numbers, signatures, template placeholders) · problem files (password-protected, corrupt, zero-byte, mislabeled)

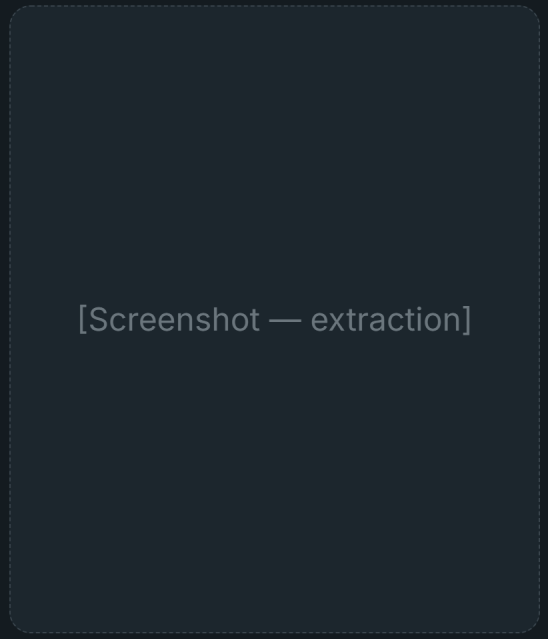
LIVE DEMO

# Ingest → exposure table → evidence drill-down → review



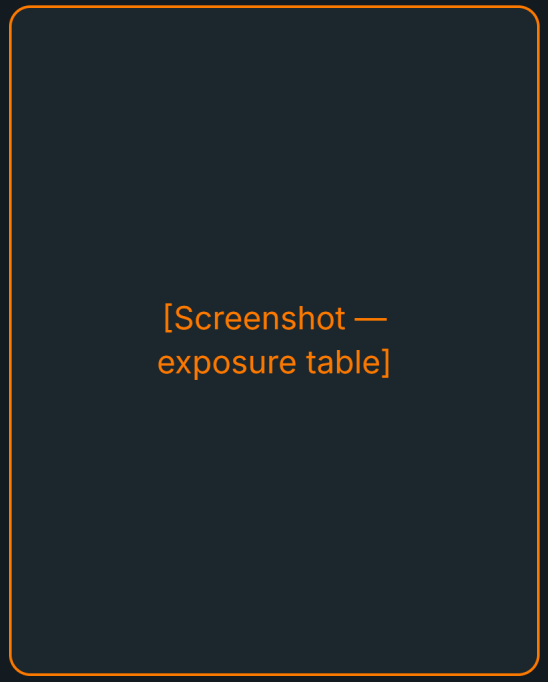

### 1. Ingest & triage

Inventory, classify, route. Nothing silently dropped — problem files quarantine with a reason.



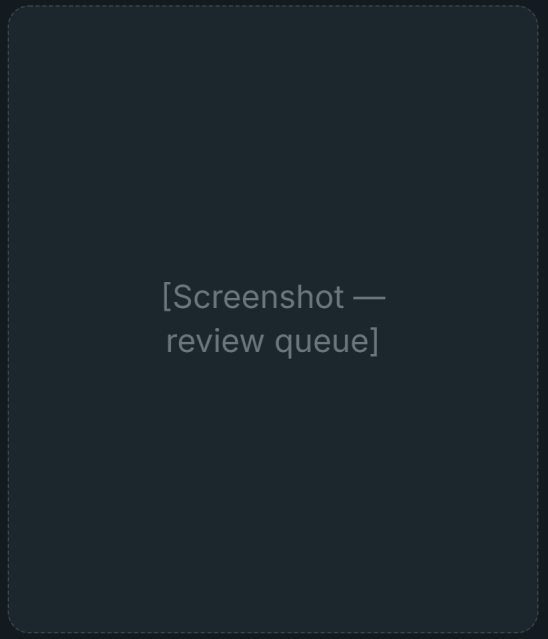

### 2. Extract & resolve

Deterministic detectors plus LLM extraction, linked to a unique individual.




### 3. Exposure table

One row per person, every flag clickable to its source passage.




### 4. Review queue

Low-confidence extractions and uncertain matches, reviewer decisions stored.

FORCED FAILURE, DEMONSTRATED LIVE

# The budget gate: caught, recovered, auditable

HIT THE WALL

**200,139**

of a 200,000-token budget

orchestrator.check_budget → BUDGET_EXCEEDED
repeated every call, run halted · 2026-09-22 21:35 UTC

RECOVERED

**600,000**

budget raised, run resumed automatically

orchestrator.check_budget → OK
resumed at 200,139 tokens · 2026-09-23 01:17 UTC

No silent failure, no lost work. The hard gate stopped spend; the operator decision is visible in the trace; the run picked up at the exact document it left off on.

ACCURACY VS. MANIFEST

# Measured against planted ground truth, not vibes

**[__%]**
overall field-extraction recall

**[__%]**
entity-linking accuracy (variant → primary)

**[__/__]**
false positives on negative sample (want 0)

**[__%]**
quarantine-reason correctness

**Weakest categories, where we focused review**

Scanned financial accounts, SSNs on scanned documents, driver licenses, passports, and name-variant matching — full per-category breakdown and error analysis in the written accuracy report.

COST ENGINEERING

# Tiered routing, measured not assumed

| Configuration | Tokens used | Tokens / doc | Recall | Time |
|---|---|---|---|---|
| Current — threshold 0.7 | [_____] | [___] | [__%] | [___s] |
| Aggressive — threshold 0.9 | [_____] | [___] | [__%] | [___s] |

Targeted comparison on a 60-document weak-category sample, not the full corpus.

**Extrapolated cost at scale**

100K documents ≈ **[$____]**  1M documents ≈ **[$_____]**

Recommended configuration: [current / aggressive], based on the recall-per-dollar trade-off above.

OBSERVABILITY

# Every step, every tool call, fully inspectable

### GET /traces/{run_id}

Live-tested against production run data. Every row: agent, tool call, input, output, timestamp.

```
orchestrator · check_budget
tokens=200139/200000 → BUDGET_EXCEEDED
2026-09-22T21:35:13Z
```

### Three required run traces

✓ Budget / approval-gate hit — captured

◦ Clean run — in progress

◦ Recovered failure — in progress

STATUS

# What's proven, what's still open

## CONFIRMED & LIVE-TESTED

- Full corpus processed end-to-end, every category
- Per-category accuracy report against manifest
- Budget-exceeded → recovery trace, live
- The run-trace endpoint, tested against real data

## IN PROGRESS

- Exception Investigator recovery paths — sandbox-tested only
- Two-config cost/accuracy comparison — completing the run
- 100K / 1M cost extrapolation — finalizing figures
- Clean-run and recovered-failure traces for the deck

# Questions

A deterministic pipeline doing the bulk of the work, agents reasoning only where judgment is required, every flag traceable, every failure recoverable and auditable.

[Your name] · [email]

DataFactZ AI Engineering Internship